

VirusTotal Integration with Wazuh

Automated Malware Detection through SIEM and Threat Intelligence Integration

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Domain: SIEM Administration & Threat Intelligence

Tools Used: Wazuh SIEM, VirusTotal API, Windows Endpoint

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Cybersecurity / SOC Lab Project

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1. Introduction

Modern security monitoring depends on the ability to detect, verify, and respond to suspicious files quickly. This project demonstrates the integration of Wazuh, an open-source Security Information and Event Management (SIEM) platform, with VirusTotal, a cloud-based threat intelligence and malware-scanning service. The goal is to automatically check file hashes generated by Wazuh's File Integrity Monitoring (FIM) module against VirusTotal's malware database, giving analysts immediate visibility into potentially malicious files on a monitored endpoint.

1.1 Objectives

- Deploy and configure a Wazuh SIEM manager and connect a Windows endpoint agent.
- Enable File Integrity Monitoring (FIM) on critical user directories.
- Integrate the VirusTotal API with Wazuh to automatically scan newly created or modified files.
- Validate the end-to-end detection workflow using a safe malware test sample (EICAR).
- Analyze the generated alerts and document the security findings.

1.2 Scope

The project is implemented in a lab environment consisting of one Wazuh manager/dashboard node and one Windows 11 endpoint running the Wazuh agent. The integration is limited to file-hash reputation checks performed through the VirusTotal public API and does not cover network-based threat intelligence or endpoint response/remediation actions.

2. Tools and Environment

Component	Details
SIEM Platform	Wazuh (Manager + Dashboard)
Threat Intelligence Source	VirusTotal Public API
Monitored Endpoint	Windows 11 Pro (agent: window_agent)

Component	Details
Monitoring Module	Wazuh Syscheck (File Integrity Monitoring)
Test Sample	EICAR standard anti-malware test file
Configuration File	ossec.conf (Wazuh Manager)

3. Implementation

3.1 Wazuh SIEM Dashboard

The Wazuh dashboard provides centralized security monitoring and visualization of events collected from connected endpoints, including agent status, alert severity distribution, and access to modules such as File Integrity Monitoring, Threat Hunting, and MITRE ATT&CK mapping.

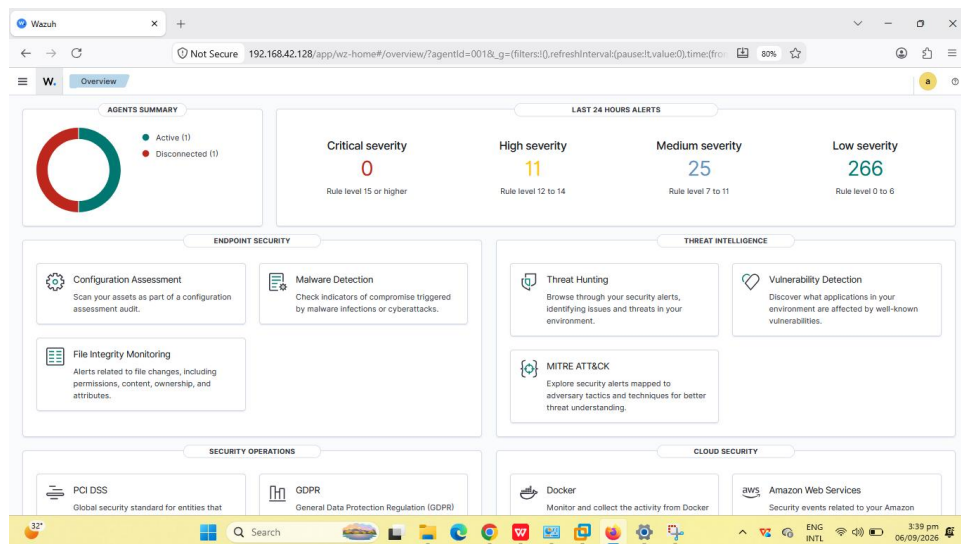


Figure 1: Wazuh dashboard overview showing agent summary and alert severity for the last 24 hours.

3.2 Installing the Wazuh Agent on Windows

The Wazuh agent is installed on the Windows endpoint to collect security events and monitor system activity. Once installed, the agent registers with the Wazuh manager and reports its status, operating system, and group membership on the dashboard.

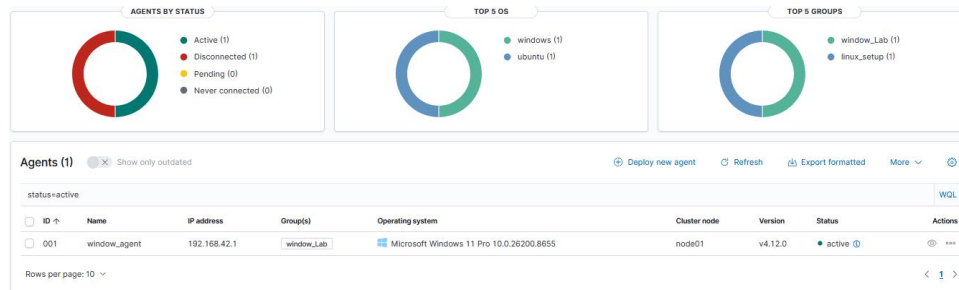


Figure 2: Registered Windows agent (*window_agent*) shown as active on the Wazuh manager.

3.3 VirusTotal API Key Configuration

A VirusTotal account is created and an API key is obtained to enable automated malware intelligence and file-hash analysis. This key authorizes the Wazuh manager to submit file hashes to VirusTotal for reputation checks.

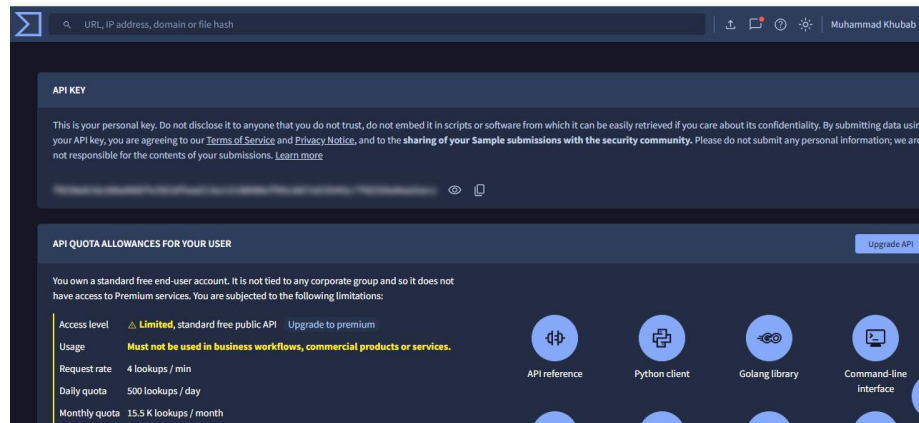


Figure 3: VirusTotal account API key page used to generate the integration key.

3.4 Wazuh–VirusTotal Integration

The VirusTotal integration is configured in the Wazuh manager's `ossec.conf` file to connect file-monitoring events with VirusTotal. The relevant configuration block is shown below:

```
<ossec_config>

  <integration>
    <name>virustotal</name>
    <api_key>your_virustotal_api_key</api_key>
    <group>syscheck</group>
    <alert_format>json</alert_format>
  </integration>
</ossec_config>
```

Note: the API key value has been replaced with a placeholder in this report and redacted in the accompanying screenshot, since it is a private credential and should never be shared or published.



Figure 4: Manager configuration panel showing the VirusTotal integration block added to ossec.conf (API key redacted).

3.5 File Integrity Monitoring Configuration

Wazuh File Integrity Monitoring (syscheck) is configured to detect newly created or modified files on the monitored endpoint. The monitored directories were set to the user's Documents and Downloads folders, which are common locations for downloaded files.

```

<!-- File integrity monitoring -->
<syscheck>
  <disabled>no</disabled>

  <!-- Frequency that syscheck is executed default every 12 hours -->
  <frequency>43200</frequency>

  <!-- Default files to be monitored. -->
  <directories check_all="yes" report_changes="yes" realtime="yes">C:\Users\M_KHUBAB\Documents</directories>
  <directories check_all="yes" report_changes="yes" realtime="yes">C:\Users\M_KHUBAB\Downloads</directories>

```

Figure 5: Syscheck configuration monitoring the Documents and Downloads directories in real time.

3.6 Malware Testing

To validate the detection workflow, a standard, harmless malware test file (EICAR) was downloaded from the official EICAR test-file page. This file is recognized by anti-malware engines as a test signature without containing any real malicious code, making it a safe way to confirm that the detection pipeline works end-to-end.

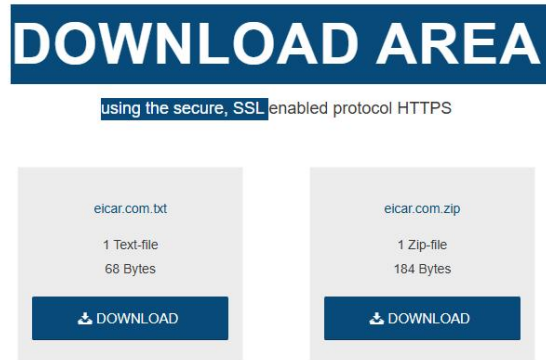


Figure 6: EICAR test-file download area used to safely trigger the detection workflow.

4. Results and Log Analysis

Wazuh logs record file-system events and VirusTotal integration results generated during the testing process. When the EICAR test file was placed in a monitored directory, Wazuh detected the file-system change and forwarded the file hash to VirusTotal, which returned a positive detection result.

228 hits						
Sep 5, 2026 @ 15:00:58.112 - Sep 6, 2026 @ 15:00:58.112						
Export Formatted 768 available fields Columns Density 1 fields sorted Full screen						
	timestamp	agent.name	rule.description	rule.level	rule.id	
	Sep 6, 2026 @ 15:00:38.015	window_agent	VirusTotal: Alert - c:\users\m_khubab\documents\leicar_com.zip - 63 engines detected t...	12	87105	
	Sep 6, 2026 @ 15:00:32.772	window_agent	VirusTotal: Alert - c:\users\m_khubab\documents\unconfirmed 639877.crdownload - 63 ...	12	87105	
	Sep 6, 2026 @ 15:00:28.351	window_agent	VirusTotal: Alert - c:\users\m_khubab\documents\unconfirmed 639877.crdownload - 63 ...	12	87105	
	Sep 6, 2026 @ 15:00:21.522	window_agent	VirusTotal: Alert - c:\users\m_khubab\downloads\b7cac479-42fa-4f12-b656-4d3f3d075...	12	87105	
	Sep 6, 2026 @ 15:00:17.414	window_agent	Integrity checksum changed.	7	550	
	Sep 6, 2026 @ 15:00:15.953	window_agent	File added to the system.	5	554	
	Sep 6, 2026 @ 15:00:15.905	window_agent	File deleted.	7	553	
	Sep 6, 2026 @ 15:00:15.903	window_agent	File added to the system.	5	554	
	Sep 6, 2026 @ 15:00:15.239	window_agent	File deleted.	7	553	
	Sep 6, 2026 @ 14:59:45.035	window_agent	VirusTotal: Alert - c:\users\m_khubab\downloads\b7cac479-42fa-4f12-b656-4d3f3d075...	12	87105	
	Sep 6, 2026 @ 14:59:35.196	window_agent	File added to the system.	5	554	
	Sep 6, 2026 @ 14:57:03.473	window_agent	VirusTotal: Alert - c:\users\m_khubab\downloads\7245209e-6fe9-4b62-96f7-71766992...	12	87105	

Figure 7: Wazuh security events showing VirusTotal alerts and file integrity events for the test endpoint.

The generated alerts were analyzed to identify the affected file, hash values, file location, file name, and detection results, giving analysts the information needed for further investigation.

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Table JSON

_index	wazuh-alerts-4.x-2026.09.06
agent.id	001
agent.ip	192.168.42.1
agent.name	window_agent
data.integration	virustotal
data.virustotal.found	1
data.virustotal.malicious	1
data.virustotal.permalink	https://www.virustotal.com/gui/file/2546dcffc5ad854d4ddc64fbf056871cd5a09f2471cb7a5bf04ac23b6e9eedad/detection/f-2546dcffc5ad854d4ddc64fbf056871cd5a09f2471cb7a5bf04ac23b6e9eedad-1788688198
data.virustotal.positives	63
data.virustotal.scan_date	2026-09-06 07:36:38
data.virustotal.sha1	d27265074c9eac2e2122ed69294dbc4d7cce9141
data.virustotal.source.alert_id	1788688815.1362981
data.virustotal.source.file	c:\users\m_khubab\documents\eicar_com.zip
data.virustotal.source.md5	6ce6f415d8475545be5ba114f288b0ff
data.virustotal.source.sha1	d27265074c9eac2e2122ed69294dbc4d7cce9141
data.virustotal.total	67
decoder.name	json
id	1788688838.1370418

Figure 8: Alert detail view showing VirusTotal scan results, including detection ratio, hashes, and permalink.

data.virustotal.total	67
decoder.name	json
id	1788688838.1370418
input.type	log
location	virustotal
manager.name	wazuh-server
rule.description	VirusTotal: Alert - c:\users\m_khubab\documents\eicar_com.zip - 63 engines detected this file
rule.firedtimes	4
rule.gdpr	IV_35.7.d
rule.groups	virustotal
rule.id	87105
rule.level	12
rule.mail	true
rule.mitre.id	T1203
rule.mitre.tactic	Execution
rule.mitre.technique	Exploitation for Client Execution
rule.pci_dss	18.6.1, 11.4
timestamp	Sep 6, 2026 @ 15:00:38.015

Figure 9: Extended alert fields showing rule classification, MITRE ATT&CK mapping (T1203), and PCI DSS/GDPR compliance tags.

4.1 Key Findings

- The test file (eicar_com.zip) was flagged by 63 out of 67 VirusTotal engines, confirming successful hash-reputation lookup.
- The alert was automatically classified at rule level 12 (high severity) and mapped to MITRE ATT&CK technique T1203 – Exploitation for Client Execution.
- Wazuh correctly logged supporting file-system events (file added / integrity checksum changed) alongside the VirusTotal alert, providing full context for the detection.

- The alert was additionally tagged against GDPR (IV_35.7.d) and PCI DSS (10.6.1, 11.4) compliance requirements, demonstrating value for compliance reporting.

5. Conclusion

The VirusTotal and Wazuh integration provides an effective approach for detecting potentially malicious files. Wazuh monitors important files and generates events when changes or new files are detected, while the file hash is analyzed through the VirusTotal API to identify known malicious files. The generated alerts provide useful information for security investigation and incident response, improving the visibility of malware-related activity in the monitored system and helping security analysts investigate suspicious files more efficiently.

Overall, this project demonstrates how SIEM and threat-intelligence capabilities can work together to strengthen malware detection, giving smaller security teams an accessible way to add automated file-reputation checks to their existing monitoring workflow.